L15

Math Exploration 1

1. The acute angles of a right triangle are $a°$ and $b°$, where $a > b$ and both a and b are prime numbers. What is the least possible value of b? (2020 AMC 10B Problems, Question 4)

A. **2** B. **3** C. **5** D. **7** E. **11**

$a + b = 90$

× 88 2

× 87 3

× 85 5

✓ 83 7

Math Exploration 2

4. A majority of the **30** students in Ms. Demeanor's class bought pencils at the school bookstore. Each of these students bought the same number of pencils, and this number was greater than **1**. The cost of a pencil in cents was greater than the number of pencils each student bought, and the total cost of all the pencils was **$17.71**. What was the cost of a pencil in cents? (2011 AMC 10A Problems, Question #10)

>15

↳ 1771

A. **7** B. **11** C. **17** D. **23** E. **77**

of students × # of pencil × unit cost = 1771

23 7 11

11 | 1771

7 | 161

23

$1771 = 7 \times 11 \times 23$

7 Among three-digit numbers, there are _____ numbers with exactly 9 factors.

Answer

7

Solution

Since $9 = 1 \times 9 = 3 \times 3$, according to the formula for the number of factors, a number with 9 factors can be represented as either a prime number raised to the power of 8, or the product of the squares of two distinct prime numbers. Among three-digit numbers, only $2^8 = 256$ satisfies the first condition. For the second condition, the numbers that satisfy the condition are $2^2 \times 5^2 = 100$, $2^2 \times 7^2 = 196$, $2^2 \times 11^2 = 484$, $2^2 \times 13^2 = 676$, $3^2 \times 5^2 = 225$, and $3^2 \times 7^2 = 441$. Therefore, there are a total of 7 numbers that satisfy the condition.

8 20! has A factors, how many factors does A have? _____

Answer

80

Solution

$20! = 2^{10+5+2+1} \times 3^{6+2} \times 5^4 \times 7^2 \times 11 \times 13 \times 17 \times 19$, so
$A = 19 \times 9 \times 5 \times 3 \times 2 \times 2 \times 2 \times 2 = 2^4 \times 3^3 \times 5 \times 19$, A has $5 \times 4 \times 2 \times 2 = 80$ factors.

9 Using $2, 3, 4, 5, 6, 7$ to form two three-digit numbers A and B, then the maximum value of greatest common factor of the three numbers A, B, and 540 could be _____ .

Answer

108

Solution

$540 = 2^2 \times 3^3 \times 5$. The greatest common divisor (GCD) of A, B, and 540 must be a factor of 540. The factors of 540 in descending order are 540, 270, 180, 135, 108, 90, and so on. Since A and B are not divisible by 10, neither 540, 270, nor 180 can be factors of both A and B. Also, since A and B cannot both be divisible by 5, 135 is not a common factor of A and B. Among the factors of 540 excluding these numbers, the largest one is 108. Considering multiples of 108 with three digits, we have 108, 216, 324, 432, 540, 648, 756, 864, 972. Among them, the numbers that can be formed using the digits $2, 3, 4, 5, 6, 7$ are 324, 432, and 756. It can be observed that when one of A and B is 756 and the other is 324 or 432, the GCD of A, B, and 540 is 108. Therefore, the largest possible GCD of A, B, and 540 is 108.

10 A box contains no more than 200 pieces of candy. If the candy is taken out in groups of 2, 3, 4, or 6 pieces at a time, exactly one piece will be left in the box each time. If the candy is taken out in groups of 11 pieces, the box will be emptied completely. There are _____ pieces of candy in the box?

Answer

121

Solution

Given that if the candies are taken out in groups of 11, the total number of candies in the box must be a multiple of 11. Since there are no more than 200 candies, the possible numbers of candies are:
$11, 22, 33, 44, 55, 66, 77, 88, 99, 110, 121, 132, 143, 154, 165, 176, 187, 198.$
Additionally, it is known that if the candies are taken out in groups of 2, 3, 4, or 6, exactly one piece will remain in the box each time. Thus, the total number of candies must be of the form $12k + 1$. The possible numbers are:
$13, 25, 37, 49, 61, 73, 85, 97, 109, 121, 133, 145, 157, 169, 181, 193.$
Taking the intersection of the two sets, we find that the only common number is 121. Therefore, the total number of candies in the box is:
$121.$

L16

Math Exploration 2

3. What is the remainder of $19^{2024} - 8^{2024}$ divided by 9?

A. 0 B. 1 C. 2 D. 4 E. 5

$19 \equiv 1 \pmod 9$

$19^{2024} \equiv 1^{2024} \pmod 9$

$\equiv 1 \pmod 9$

$8 \equiv -1 \pmod 9$

$8^{2024} \equiv (-1)^{2024} \pmod 9$

$\equiv 1 \pmod 9$

$19^{2024} - 8^{2024} \equiv 1 - 1 \equiv 0 \pmod 9$

Math Exploration 2

5. Find the reaminder when $2025 \cdot 15^{2024}$ is divided by 4?

A. 0 B. 1 C. 2 D. 3 E. 4

$2025 \equiv 1 \pmod 4$

$15 \equiv 3 \equiv -1 \pmod 4$

$15^{2024} \equiv 3^{2024} \equiv (-1)^{2024} \pmod 4$

$\equiv 1 \pmod 4$

$2025 \cdot 15^{2024} \equiv 1 \cdot 1 \equiv 1 \pmod 4$

2 There are a number of eggs in the box. If we take 3 each time, 1 is left in the end; if 5 are taken each time, 2 are left in the end; if we take 7 each time, 3 are left in the end, then there are at least _____ eggs in the box.

Answer

52

Solution

Numbers that leave a remainder of 3 when divided by 7 include: $10, 17, 24, 31, 38, 45, 52, 59$, and so on.
Among these numbers, numbers that leave a remainder of 1 when divided by 3 are: $10, 31, 52$, and so on.
Among these numbers, the only number that leaves a remainder of 2 when divided by 5 is 52.
Therefore, 52 is the smallest positive integer that satisfies all the conditions.

10 What is the remainder when $8 + 8 \times 8 + 8 \times 8 \times 8 + \cdots + \underbrace{8 \times 8 \times \cdots \times 8}_{2017}$ is dividing by 5?

A. 0 B. 1 C. 2 D. 3 E. 4

Answer

E

Solution

Since $2008 \div 7$ leaves a remainder of 6, we have $2008^n \equiv 6^n \pmod 7$.
For $6^1 \div 7$, the remainder is 6; for $6^2 \div 7$, the remainder is 1; and for $6^3 \div 7$, the remainder is 6.
Thus, $6^{2008} \div 7$ also leaves a remainder of 1, meaning $2008^{2008} \div 7$ leaves a remainder of 1.
Therefore, in 2008^{2008} days, it will be Sunday.

L17

Math Exploration 1

2. A dessert chef prepares the dessert for every day of a week starting with Sunday. The dessert each day is either cake, pie, ice cream, or pudding. The same dessert may not be served two days in a row. There must be cake on Friday because of a birthday. How many different dessert menus for the week are possible? (2012 AMC 10B Problems, Question #11)

A. 729 B. 972 C. 1024 D. 2187 E. 2304

Sun Mon Tues Wed Thurs Fri Sat

$_3C_1$ $_3C_1$ $_3C_1$ $_3C_1$ $_3C_1$ cake $_3C_1$

$(_3C_1)^6 = 3^6 = 729$

Math Exploration 2

$1+2+3+\cdots+9+1 = 46$

2. The number **2013** has the property that its units digit is the sum of its other digits, that is $2+0+1=3$. How many integers less than **2013** but greater than **1000** share this property? (2013 AMC10B Problems, Question #18)

A. 33 B. 34 C. 45 D. 46 E. 58

From $2000 \le N < 2013$

2002

From $1000 < N \le 1999$

① 1001

② 1102, 01

③ 1023 20, 11

④ 1214 12, 30, 03

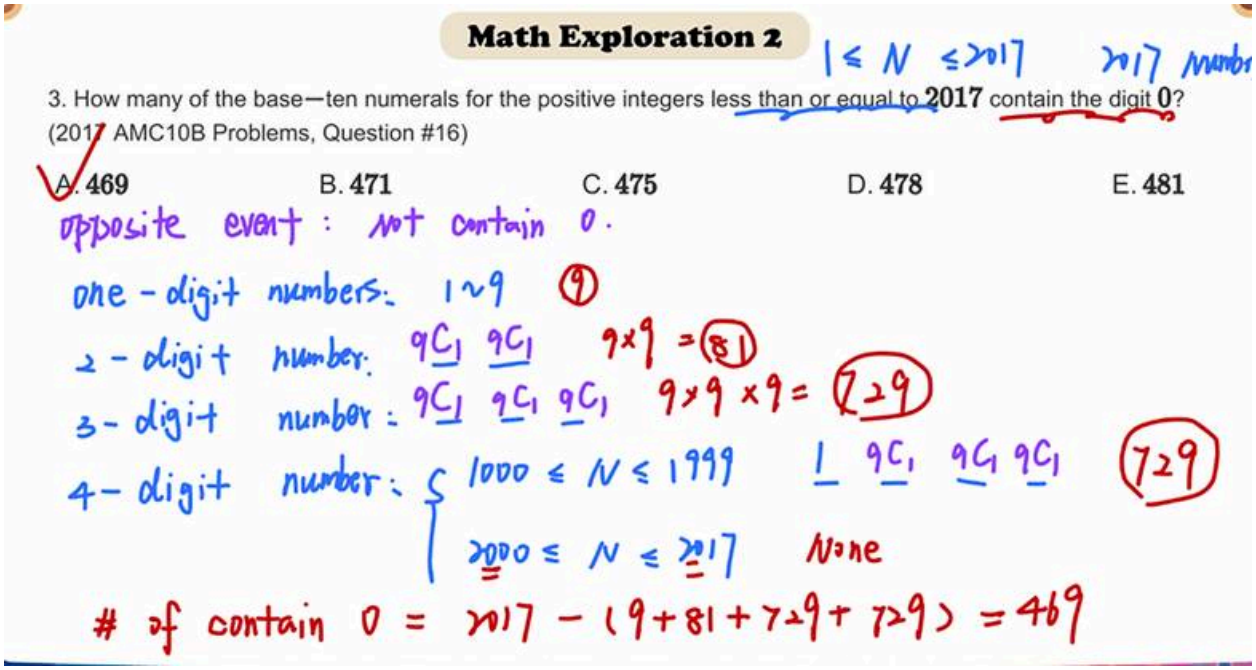
⑤ 1225 13, 04, 31, 40

⑥ 1056 14, 23, 32, 41, 50

⑦ 1337 42, 24, 15, 51, 60, 06

⑧ 1348 43, 52, 25, 61, 16, 07, 70

⑨ 1089 80, 17, 71, 26, 62, 35, 53, 44



1 The side length of the three sides of a triangle is an integer, and the longest side length is 7 cm. How many types of triangles are there? _____

Answer

16

Solution

Let the other two sides have lengths of a and b, and $a \geqslant b$, then we have :

$$\begin{cases} 1 \leqslant a \leqslant 7 \\ 1 \leqslant b \leqslant 7 \\ a \geqslant b \\ a+b \geqslant 8 \\ a-b \leqslant 6 \end{cases},$$

After enumeration, we can get:

$(b, a) = (1,7)\ (2,6)\ (2,7)\ (3,5)\ (3,6)\ (3,7)\ (4,4)\ (4,5)\ (4,6)\ (4,7)\ (5,5)\ (5,6)\ (5,7)\ (6,6)\ (6,7)\ (7,7)$

So there are 16 combinations.

2 The sum of the digits on each digit of a three-digit odd number is 11. How many such three digits are there? _____

29

Solution

Method 1:
Unit digit is 1: $911/821/731/641/551/461/371/281/191$;
Unit digit is 3: $803/713/623/533/443/353/263/173$;
Unit digit is 5: $605/515/425/335/245/155$;
Unit digit is 7: $407/317/227/137$;
Unit digit is 9: $209/119$;
total: $9+8+6+4+2=29$.
Method 2:
his problem can also be solved by dictionary arrangement.
$119, 137, 155, 173, 191$;
$209, 227, 245, 263, 281$;
$317, 335, 353, 371$;
$407, 425, 443, 461$;
$515, 533, 551$;
$605, 623, 641$;
$713, 731$;
$803, 821$;
911;
Total: $5\times 2+4\times 2+3\times 2+2\times 2+1=29$ such number.

(2) 6 students line up, A and B must be next to each other, B and C cannot be next to each other. There are _____ ways to arrange.

(2) Exclusion: $\mathrm{A}_2^2\times\mathrm{A}_5^5-\mathrm{A}_2^2\times\mathrm{A}_4^4=192$.

4 Find the number of different permutations:

(4) 4 men and 4 women stand in a line. People of the same sex cannot be next to each other.

(4) It is a 'non-adjacent' problem, and we can use the insertion method to solve: $\mathrm{A}_4^4\mathrm{A}_4^4\mathrm{A}_2^1=1152$.

5 Seven students took pictures and found out how many standing methods there were under the following conditions:

(2) Seven people are lined up, and two classmates can't stand on the side. How many ways to stand?

(3) Tim and Bob have at least one person standing on the side. How many ways to stand?

(2) The left and right ends are two special locations, so both positions need to be prioritized. First, choose 2 of the 5 people other than the two classmates who cannot stand on the side, and then arrange the remaining 5 people, so there are $A_5^2 \times A_5^5 = 2400$ arrangements.

(3) "At least" suggests that we can think from the opposite, i.e., exclusion. Tim and Bob have at least one person standing on the side, so the opposite is: neither of them is on the side, so the elimination method only needs to exclude the situation in (2), so we have $A_7^7 - A_5^2 \times A_5^5 = 2640$ ways.

6 Among the 10 students, 5 can install computers, 3 of them can install audio equipment, and the remaining 2 can install both computers and audio equipment. Today, the school decided to dispatch a 6-people installation team. In this team, 3 people are required to install computers, and 3 people are required to install audio equipment, so there are _____ different selection schemes.

Answer

185

Solution

Categorized by students with dual skills:

(1) neither of them will be dispatched, there are $C_5^3 = \frac{5 \times 4 \times 3}{3 \times 2 \times 1} = 10$ ways;

(2) dispatch one. Then we have 2 choices. The tasks for this person are divided into two categories:

If the person installs a computer, 2 more person needs to install a computer, so there are $C_5^2 = \frac{5 \times 4}{2 \times 1} = 10$ ways, and the 3 people who will install audio equipment are all selected, and there are only 1 selection methods. By multiplication principle, there are $10 \times 1 = 10$ ways.

If the person installs audio equipment, then we need to choose 2 people from the 3 people who can install audio equipment. Then there are $C_3^2 = \frac{3 \times 2}{2 \times 1} = 3$ ways. Then we need to choose 3 poeple from the 5 people who can install computer, so there are $C_5^3 = \frac{5 \times 4 \times 3}{3 \times 2 \times 1} = 10$ ways. Then by multiplication principle, there are $3 \times 10 = 30$ ways.

By summation principle, there are $10 + 30 = 40$ ways.

Therefore, there are $2 \times 40 = 80$ ways in total.

(3) dispatch both of them, the task for these two people can be categorized and discussd as follows:

① Both of them will install computer, then we still need to choose 1 person from the 5 people who can install computer, then select the three people who can install audio equipment. So we have $5 \times 1 = 5$ ways to dispatch.

② One of these two people will install computer, and the other will install audio equipment, these two can be exchange, so we can $C_5^2 \times C_3^2 \times 2 = \frac{5 \times 4}{2 \times 1} \times \frac{3 \times 2}{2 \times 1} \times 2 = 60$ ways to dispatch;

③ Both of them will install audio equipment, then there are $3 \times C_5^3 = 3 \times \frac{5 \times 4 \times 3}{3 \times 2 \times 1} = 30$ ways to dispatch.

According to the summation principle, there are $5 + 60 + 30 = 95$ ways to dispatch.

Therefore, there are $90 + 95 = 185$ ways.

8 12 identical apples should be divided into 3 days to eat, at least 2 on the first day, allow to eat no apple on the second day, and at least 3 on the third day, then there are _____ ways to eat.

Answer

36

Solution

Insertion method: give 1 on the first day, 2 on the third day, and then lend 1 on the next day. So that it is equivalent to $12-1-2+1=10$ apples eaten in 3 days at least 1 apple per day. So we have $\mathrm{C}_9^2=\dfrac{9\times 8}{2\times 1}=36$ ways.

L18

Math Exploration 2

Ms. Carr asks her students to read any **5** of the **10** books on a reading list. Harold randomly selects **5** books from this list, and Betty does the same. What is the probability that there are exactly **2** books that they both select? (2020 AMC 10B Problems, Question #11)

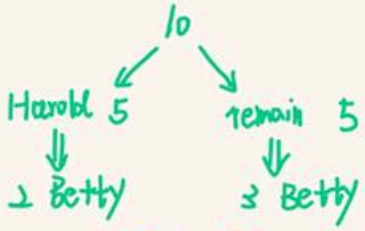
A. $\frac{1}{8}$ B. $\frac{5}{36}$ C. $\frac{14}{45}$ D. $\frac{25}{63}$ E. $\frac{1}{2}$

Total outcomes $= {}_{10}C_5 \cdot {}_{10}C_5$

Step 1: Harold choose 5 books: ${}_{10}C_5$

Step 2: 2 book both select
$\Rightarrow$ Betty 2 books from Harold. ${}_5C_2$

Step 3: Betty 3 books from remaining ${}_5C_3$



$$P = \frac{{}_{10}C_5 \cdot {}_5C_2 \cdot {}_5C_3}{{}_{10}C_5 \cdot {}_{10}C_5} = \frac{25}{63}$$

Math Exploration 2

An envelope contains eight bills: **2** ones, **2** fives, **2** tens, and **2** twenties. Two bills are drawn at random without replacement. What is the probability that their sum is **$20** or more? (2005 AMC 12B Problems, Question #11)

A. $\frac{1}{4}$ B. $\frac{2}{7}$ C. $\frac{3}{7}$ D. $\frac{1}{2}$ E. $\frac{2}{3}$

Total outcomes $= {}_8C_2 = \frac{8 \times 7}{2} = 28$

Case 1: sum $= 20 = 10 + 10$ 1 way

Case 2: sum > 20

if 1 twenty, ${}_2C_1 \cdot {}_6C_1 = 12$ ways.

if 2 twenties. 1 way

Desired outcomes $= 1 + 12 + 1 = 14$ $P = \frac{14}{28} = \frac{1}{2}$

Math Exploration 3

A box contains **5** chips, numbered **1**, **2**, **3**, **4**, and **5**. Chips are drawn randomly one at a time without replacement until the sum of the values drawn exceeds **4**. What is the probability that **3** draws are required? (2018 AMC 10B Problems, Question #6)

A. $\frac{1}{15}$ B. $\frac{1}{10}$ C. $\frac{1}{6}$ D. $\frac{1}{5}$ E. $\frac{1}{4}$

Total outcomes $= {}_5C_2 = 10$

Desired outcomes $= 2$

$1+2=3$

$1+3=4$

$P = \frac{2}{10} = \frac{1}{5}$

Math Exploration 3

For a particular peculiar pair of dice, the probabilities of rolling **1**, **2**, **3**, **4**, **5** and **6** on each die are in the ratio **1 : 2 : 3 : 4 : 5 : 6** What is the probability of rolling a total of **7** on the two dice? (2006 AMC 12B Problems, Question #17)

A. $\frac{4}{63}$ B. $\frac{1}{8}$ C. $\frac{8}{63}$ D. $\frac{1}{6}$ E. $\frac{2}{7}$

Assume $p + 2p + 3p + 4p + 5p + 6p = 1$ $\quad p = \frac{1}{21}$

$P(1) = \frac{1}{21}$, $P(2) = \frac{2}{21}$, $P(3) = \frac{3}{21}$, $P(4) = \frac{4}{21}$, $P(5) = \frac{5}{21}$, $P(6) = \frac{6}{21}$

(1,6) (6,1) $P = \frac{1}{21} \times \frac{6}{21} \times 2$

(2,5) (5,2) $P = \frac{2}{21} \times \frac{5}{21} \times 2$

(3,4) (4,3) $P = \frac{3}{21} \times \frac{4}{21} \times 2$

$\Rightarrow P = \frac{8}{63}$

Math Exploration 3

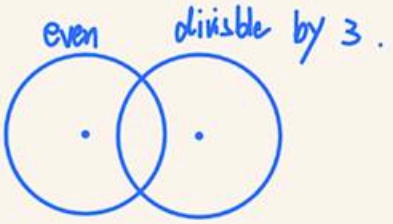
A number is randomly selected from $1, 2, 3, \ldots, 30$. What is the probability that it is an even number or divisible by 3?

A. $\frac{1}{6}$ B. $\frac{1}{3}$ C. $\frac{1}{2}$ D. $\frac{2}{3}$ ✓

$P(\text{even}) = \frac{1}{2}$

$P(\text{divisible by } 3) = \frac{1}{3}$

$P(\text{even and divisible by } 3) = \frac{5}{30} = \frac{1}{6}$



$P = \frac{1}{2} + \frac{1}{3} - \frac{1}{6} = \frac{2}{3}$

Math Exploration 4

Una rolls **6** standard **6**-sided dice simultaneously and calculates the product of the **6** numbers obtained. What is the probability that the product is divisible by **4**?(2021 Fall AMC 10B Problems, Question #14)

⇒ Complementary = Not divisible by 4

A. $\frac{3}{4}$ B. $\frac{57}{64}$ C. $\frac{59}{64}$ ✓ D. $\frac{187}{192}$ E. $\frac{63}{64}$

Case 1: 6 odd numbers.

$P(6 \text{ odd}) = (\frac{1}{2})^6 = \frac{1}{64}$

Case 2: 5 odd numbers, only 1 even numbers (2 or 6)

$P(5 \text{ odd and } 2/6) = (\frac{1}{2})^5 \cdot \frac{1}{3} \cdot 6 = \frac{1}{16}$

$P(\text{divisible by } 4) = 1 - (\frac{1}{64} + \frac{1}{16}) = \frac{59}{64}$

3 Five students A, B, C, D and E stand in a row. The probability that A and B are not adjacent to each other is _____ .

A $\frac{4}{5}$ B. $\frac{3}{5}$ C. $\frac{2}{5}$ D. $\frac{1}{5}$

B

Solution

Total number of independent events is $n = {}_5P_5 = 120$,
A and B not adjacent to each other:
$m = {}_3P_3 \cdot {}_4P_2 = 72$,
Thus $P = \frac{m}{n} = \frac{72}{120} = \frac{3}{5}$.

6 If we pick two different numbers from $-2, -1, 0, 3, 4; 5$ and multiply them up, the probability that the product is a multiple of 5 is _____ .

A. $\frac{2}{5}$ B. $\frac{3}{5}$ C. $\frac{1}{4}$ D. $\frac{1}{3}$ E. $\frac{1}{2}$

Answer

B

Solution

The condition is satisfied if the product is 0, and this only happens when one of the numbers is 0. There are five ways of the second number. If the product is not 0, there are 4 ways.There are in total $\left(C_2^6\right)$ ways to choose two numbers, so the probability is $\frac{9}{15} = \frac{3}{5}$.

7 Holly writes down all the two-digit numbers which can be formed using the digits 1, 3, 7, and 9 (including 11, 33, 77 and 99). Warren selects one of these numbers at random. What is the probability that a prime number is selected?

A. $\frac{5}{8}$ B. $\frac{1}{2}$ C. $\frac{9}{16}$ D. $\frac{11}{16}$ E. $\frac{3}{4}$

Answer

A

Solution

There are $4 \times 4 = 16$ numbers in total. There are 10 prime numbers (11, 13, 17, 19, 31, 37, 71, 73, 79, 97). Therefore, the probability is $\frac{10}{16} = \frac{5}{8}$.

9 Roll the dice twice. Let the numbers on the face be m and n. What is the probability of the equation: $x^2 + mx + n = 0$ has integer solutions?

A. $\frac{5}{36}$ B. $\frac{7}{36}$ C. $\frac{1}{6}$ D. $\frac{5}{18}$ E. $\frac{1}{3}$

Answer

B

Solution

For each possible value of m (from rolling the first die), we need to check how many possible values of n (from rolling the second die) would make $m^2 - 4n$ a perfect square.
On checking all possibilities, we find that there are 7 combinations (m, n) that satisfy this condition: $(2, 1), (3, 2), (4, 3), (4, 4), (5, 4), (5, 6)$, and $(6, 5)$. Since there are a total of 36 possible outcomes when rolling two dice, the probability is thus $\frac{7}{36}$. So the correct answer is B, $\frac{7}{36}$.

10 If a certain type of cell cannot divide, it dies. The probability of a cell either dying or dividing into two cells is $\frac{1}{2}$. Starting with two such cells, what is the probability that there are still living cells after two rounds of division?

A. $\frac{39}{64}$ B. $\frac{25}{64}$ C. $\frac{31}{64}$ D. $\frac{29}{64}$

Answer

A

Solution

The probability of a single cell surviving after two rounds of division is $\frac{1}{2} \cdot \left(1 - \frac{1}{2} \cdot \frac{1}{2}\right) = \frac{3}{8}$.

Therefore, the probability of having at least one living cell after two rounds of division with two cells is $1 - \left(1 - \frac{3}{8}\right)^2 = \frac{39}{64}$.

L19

Math Exploration 1

2. A cylindrical container is put horizontally. The radius of the cross section is **1m**. If the depth of the water in the container is **0.4m**, then the width of the water is _____ .

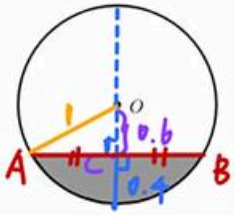
A. **0.8m** B. **1.2m** C. **1.6m** D. **1.8m**

$OC = 0.6$ $AC = BC = \frac{1}{2}AB$

Rt$\triangle AOC$, $AO = 1$, $OC = 0.6$

$AC = \sqrt{1^2 - 0.6^2} = \sqrt{0.64} = 0.8$

$AB = 1.6$



Math Exploration 2

2. As shown in the figure, in $\odot O$, chord $\overline{BC}$ is parallel to $\overline{OA}$. $\overline{AC}$ and $\overline{OB}$ intersect at M. $m\angle C = 20°$. Then $m\angle AMB$ is _____ .

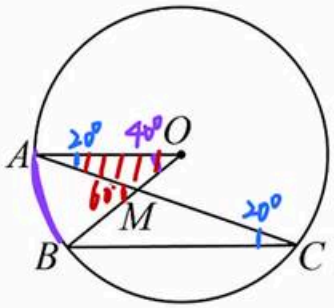
A. **30°** B. **60°** C. **50°** D. **40°**

$m\angle OAC = 20° = m\angle C$

$m\angle C$ corresponds to $\overset{\frown}{AB} \Rightarrow m\angle AOB$

$m\angle C = 20°$, $m\angle AOB = 40°$

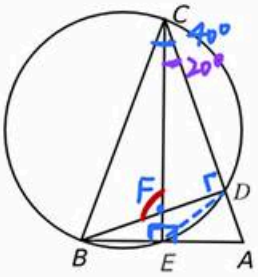
$m\angle AMB = 20° + 40° = 60°$



Math Exploration 3

4. Let $\triangle ABC$ be an isosceles triangle with $BC = AC$ and $m\angle ACB = 40°$. Construct the circle with diameter $\overline{BC}$, and let D and E be the other intersection points of the circle with the sides $\overline{AC}$ and $\overline{AB}$, respectively. Let F be the intersection of the diagonals of the quadrilateral $BCDE$. What is the degree measure of $\angle BFC$? (2019 AMC 10A Problem, Question #13)

A. 90 B. 100 C. 105 D. 110 E. 120



isos $\triangle ABC$, $m\angle CEB = 90°$, $\overline{CE} \perp \overline{AB}$, $m\angle ACE = 20°$

$m\angle BFC = 20 + 90 = 110°$

1 The number of incorrect statements is _____ .

①The major arc must be longer than the minor arc;

②Two circles with the same area are congruent circles;

③The arcs with the same length are congruent arcs;

④Through a point inside the circle, we can draw infinite chords.

⑤Through a point inside the circle, we can draw infinite diameters.

A 1 B. 2 C. 3 D. 4

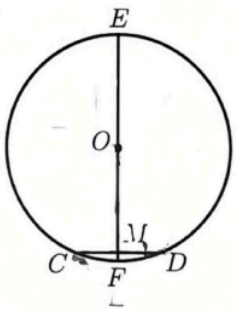
Answer

C

Solution

① Incorrect: the major arc is not necessarily longer than the minor arc, for example: the minor arc of a circle with a larger radius may be longer than the major arc with a smaller radius;
② Correct: the area is equal, then the radius is equal;
③ Incorrect: the same arc or equal arc precondition should be added with the same circle or congruent circle;
④ Correct;
⑤ Incorrect: if this point is not the center of the circle, only one diameter can be drawn. If this point is the center of the circle, an infinite number of diameters can be drawn.
So choose C.

4 As shown in the figure, in $\odot O$, diameter $\overline{EF} \perp \overline{CD}$ at point M. If $CD = 2$, and $EM = 5$, then the radius of $\odot O$ is _____ .

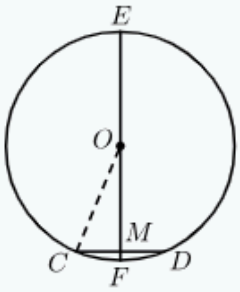


Answer

2.6

Solution

As shown in the figure, draw $\overline{CO}$:



since diameter $\overline{EF} \perp \overline{CD}$, and $CD = 2$,

$CM = \frac{1}{2}CD = \frac{1}{2} \times 2 = 1.$

Let the radius of $\odot O$ be r;

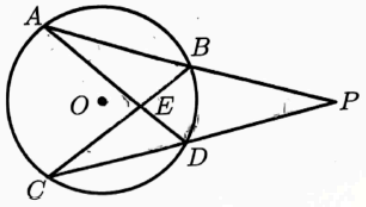
in $\mathrm{Rt}\triangle CMO$, $CM^2 + MO^2 = CO^2$,

so $1^2 + (5 - r)^2 = r^2$.

Solve to get that $r = 2.6$,

so the radius of $\odot O$ is 2.6.

9 As shown in the figure below, $\overline{AB}$ and $\overline{CD}$ are two chords of $\odot O$. $\overrightarrow{AB}$ and $\overrightarrow{CD}$ intersect at P. Connect $\overline{AD}$, $\overline{BC}$ and they intersect at E, $m\angle P = 30°$, $m\angle ABC = 50°$. Find $m\angle A$ and $m\angle AEC$.



Answer

$m\angle A = 20°$, $m\angle AEC = 70°$.

Solution

Since $\overset{\frown}{AC} = \overset{\frown}{AC}$,
we have $m\angle ABC = m\angle ADC = 50°$.
Since $m\angle ADC = m\angle A + m\angle P$,
then $m\angle P = 30°$.
So $m\angle A = m\angle ADC - m\angle P$,
$= 50° - 30°$,
$= 20°$.
$m\angle AEC = m\angle A + m\angle ABC$,
$= 20° + 50°$
$= 70°$.
Thus $m\angle A = 20°$, $m\angle AEC = 70°$.

L20

Math Exploration 1

2. On the number line, number represented by point A is 3 and number represented by point B is a. The radius of $\odot A$ is 2. Which of the following is incorrect?

A. If $a < 5$, point B is inside $\odot A$. $a<1$

B. If $1 < a < 5$, point B is inside $\odot A$.

C. If $a < 1$, point B is outside $\odot A$.

D. If $a > 5$, point B is outside $\odot A$.



Math Exploration 2

1. On the coordinate plane, circle O is centered at $(-3, 4)$ and the radius of it is 4, then _____ .

A. x–axis intersects the circle at two points and y–axis is tangent to the circle.

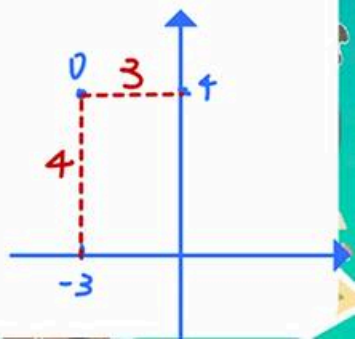
B. x–axis misses the circle and y–axis intersects the circle at two points.

C. x–axis is tangent to the circle and y–axis intersects the circle at two points.

D. x–axis is tangent to the circle and y–axis misses the circle.

$d_{O \to x\text{-axis}} = 4 = 4$, $d = r$. tangent

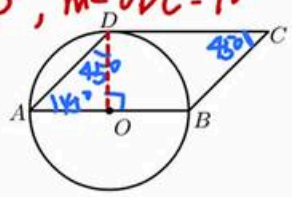
$d_{O \to y\text{-axis}} = 3 < 4$, $d < r$ intersect



Math Exploration 3

2. As shown in the figure below, quadrilateral $ABCD$ is a parallelogram. $m\angle C = 45°$, Draw $\odot O$ with $\overline{AB}$ being the diameter and it passes through D. Prove: $\overline{CD}$ is tangent to $\odot O$.

Prove $\overline{CD} \perp \overline{OD}$, $m\angle ODC = 90°$



$m\angle C = m\angle A = 45°$

$OA = OD =$ radius,

$\triangle AOD$ iso $\triangle$. $m\angle A = m\angle ADO = 45°$

$m\angle AOD = 90°$

$CD \parallel AB$, $m\angle ODC = 90°$, $\overline{CD} \perp \overline{OD}$, $\overline{CD}$ tangent.

Math Exploration 5

2. As shown in the figure, $\overline{PA}$ and $\overline{PB}$ are tangent to the circle O at points A and B, respectively. $\overline{EF}$ is tangent to circle O and intersects $\overline{PA}$ and $\overline{PB}$ at points E and F, respectively. The tangent point C lies on the arc AB. If the length of PA is 2, then the perimeter of triangle PEF is 4.

$\overline{PA} \perp \overline{AO}$, $\overline{PB} \perp \overline{BO}$

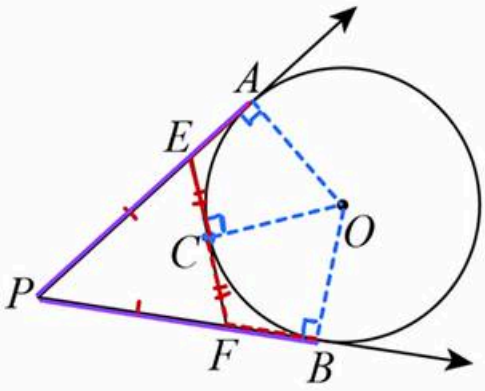
$EC = EA$, $FC = FB$

Perimeter $= PE + PF + EF$

$= PE + PF + EC + CF$

$= PE + PF + EA + FB$

$= 2 + 2 = 4$



Given that the radius of $\odot O$ is 4, and $OP = 3$, then the positional relationship between P and $\odot O$ is _____ .

A. P is inside the circle
B. P is on the circle
C. P is outside the circle
D. Cannot be determined

Answer

A

Solution

Since $OP = 3 < 4$,
the positional relationship between P and $\odot O$ is that P is inside the circle.

4 In right triangle $\triangle ABC$, $m\angle C = 90°$, $AC = 4$, and $BC = 3$. Then the positional relationship between $\odot C$ whose radius is 2.5 and $\overleftrightarrow{AB}$ is _____

Answer

$\overleftrightarrow{AB}$ intersects $\odot C$ at two points

Solution

In right triangle ABC,
$m\angle C = 90°$, $AC = 4$, and $BC = 3$.
So $AB = 5$.
Since the distance from C to $\overleftrightarrow{AB}$ is $h = \frac{3 \times 4}{5} = 2.4 < 2.5$, $\overleftrightarrow{AB}$ intersects $\odot C$ at two points.

5 Line l intersects $\odot O$. The radius of $\odot O$ is r, and the distance from point O to line l is 6. Then the range of r is _____ .

A. $r < 6$ B. $r = 6$ C. $r > 6$ D. $r \geqslant 6$

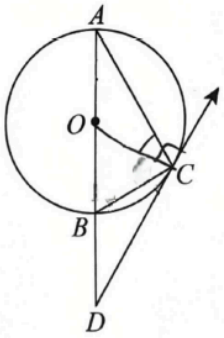
Answer

D

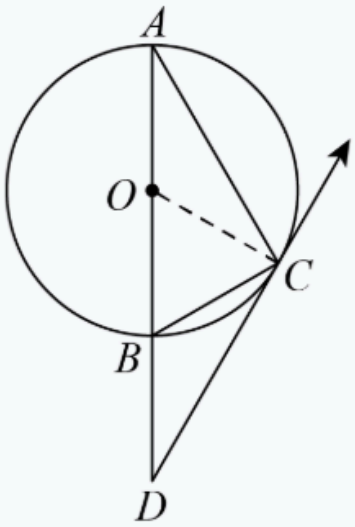
Solution

Let the distance from the center to l be d;
when $d > r$, the line is the exterior of the circle;
when $d = r$, the line is tangent to the circle;
when $d < r$, the line is a secant of the circle.
And vice versa, which means that when the line intersects the circle at two points or line l is tangent to the circle, $r \geqslant 6$.
Therefore the correct answer is D.

6 As shown in the figure, $\overline{AB}$ is the diameter of $\odot O$. C is on $\odot O$, D is on $\overrightarrow{AB}$, and $m\angle DCB = m\angle A$. Prove that $\overline{CD}$ is tangent to $\odot O$.



Draw $\overline{OC}$.



Since $\overline{AB}$ is the diameter of $\odot O$,
$m\angle ACB = 90°$,
so $m\angle A + m\angle ABC = 90°$.
Since $OB = OC$,
$m\angle OBC = m\angle OCB$.
Since $m\angle DCB = m\angle A$,
$m\angle A + m\angle ABC = m\angle DCB + m\angle OCB = 90°$.
Therefore $\overline{OC} \perp \overline{DC}$,
so $\overline{CD}$ is tangent to $\odot O$.